

Decision Tree Using Entropy and Information Gain

Play Tennis dataset - detailed pen-and-paper method

Goal: choose the attribute with the highest Information Gain at each step. Entropy measures impurity or uncertainty in the data.

1. Play Tennis Dataset

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

2. Formulae Used

$$\text{Entropy}(S) = - p(\text{Yes}) \log_2 p(\text{Yes}) - p(\text{No}) \log_2 p(\text{No})$$

$$\text{Information Gain, IG}(S, A) = \text{Entropy}(S) - \text{Weighted Entropy}(A)$$

$$\text{Weighted Entropy}(A) = \sum [|S_v| / |S|] * \text{Entropy}(S_v)$$

Here, S is the current dataset and S_v is one subset produced after splitting on attribute A .

3. Entropy of the Whole Dataset

Count the target values in the full dataset:

Yes = 9
No = 5
Total = 14

Substitute in the entropy formula:

$$\begin{aligned}\text{Entropy}(S) &= - (9/14) \log_2(9/14) - (5/14) \log_2(5/14) \\ &= - (0.643) \log_2(0.643) - (0.357) \log_2(0.357) \\ &= - (0.643)(-0.637) - (0.357)(-1.485) \\ &= 0.4096 + 0.5301\end{aligned}$$

$$= 0.9397$$

$$= 0.940$$

Therefore, Entropy(S) = 0.940

4. Compute Information Gain for Each Attribute

We now compute the weighted entropy for each attribute and subtract it from 0.940.

4.1 Attribute: Outlook

Possible values: Sunny, Overcast, Rain

Sunny subset -> rows 1, 2, 8, 9, 11

Yes = 2, No = 3

$$\begin{aligned}\text{Entropy}(\text{Sunny}) &= - (2/5) \log_2(2/5) - (3/5) \log_2(3/5) \\ &= - (0.4)(-1.322) - (0.6)(-0.737) \\ &= 0.5288 + 0.4422 \\ &= 0.971\end{aligned}$$

Overcast subset -> rows 3, 7, 12, 13

Yes = 4, No = 0

$$\begin{aligned}\text{Entropy}(\text{Overcast}) &= - (4/4) \log_2(4/4) - (0/4) \log_2(0/4) \\ &= - (1) \log_2(1) - 0 \\ &= 0\end{aligned}$$

Rain subset -> rows 4, 5, 6, 10, 14

Yes = 3, No = 2

$$\begin{aligned}\text{Entropy}(\text{Rain}) &= - (3/5) \log_2(3/5) - (2/5) \log_2(2/5) \\ &= - (0.6)(-0.737) - (0.4)(-1.322) \\ &= 0.4422 + 0.5288 \\ &= 0.971\end{aligned}$$

Weighted Entropy(Outlook)

$$\begin{aligned}&= (5/14)(0.971) + (4/14)(0) + (5/14)(0.971) \\ &= 0.3468 + 0 + 0.3468 \\ &= 0.6936 \\ &= 0.694\end{aligned}$$

$$\begin{aligned}\text{IG}(\text{Outlook}) &= \text{Entropy}(S) - \text{Weighted Entropy}(\text{Outlook}) \\ &= 0.940 - 0.694 \\ &= 0.246\end{aligned}$$

4.2 Attribute: Temperature

Hot subset -> rows 1, 2, 3, 13

Yes = 2, No = 2

$$\begin{aligned}\text{Entropy}(\text{Hot}) &= - (2/4) \log_2(2/4) - (2/4) \log_2(2/4) \\ &= - (0.5)(-1) - (0.5)(-1) \\ &= 0.5 + 0.5 \\ &= 1.000\end{aligned}$$

Mild subset -> rows 4, 8, 10, 11, 12, 14

Yes = 4, No = 2

$$\begin{aligned}\text{Entropy}(\text{Mild}) &= - (4/6) \log_2(4/6) - (2/6) \log_2(2/6) \\ &= - (0.667)(-0.585) - (0.333)(-1.585) \\ &= 0.3902 + 0.5278 \\ &= 0.918\end{aligned}$$

Cool subset -> rows 5, 6, 7, 9

Yes = 3, No = 1

$$\begin{aligned}\text{Entropy}(\text{Cool}) &= - (3/4) \log_2(3/4) - (1/4) \log_2(1/4) \\ &= - (0.75)(-0.415) - (0.25)(-2) \\ &= 0.3113 + 0.5000 \\ &= 0.811\end{aligned}$$

Weighted Entropy(Temperature)
 $= (4/14)(1.000) + (6/14)(0.918) + (4/14)(0.811)$
 $= 0.2857 + 0.3934 + 0.2317$
 $= 0.9108$
 $= 0.911$

IG(Temperature) = 0.940 - 0.911 = 0.029

4.3 Attribute: Humidity

High subset -> rows 1, 2, 3, 4, 8, 12, 14
 Yes = 3, No = 4
 Entropy(High) = $- (3/7) \log_2(3/7) - (4/7) \log_2(4/7)$
 $= - (0.429)(-1.222) - (0.571)(-0.807)$
 $= 0.5242 + 0.4608$
 $= 0.985$

Normal subset -> rows 5, 6, 7, 9, 10, 11, 13
 Yes = 6, No = 1
 Entropy(Normal) = $- (6/7) \log_2(6/7) - (1/7) \log_2(1/7)$
 $= - (0.857)(-0.222) - (0.143)(-2.807)$
 $= 0.1903 + 0.4013$
 $= 0.5916$
 $= 0.592$

Weighted Entropy(Humidity)
 $= (7/14)(0.985) + (7/14)(0.592)$
 $= 0.4925 + 0.2960$
 $= 0.7885$
 $= 0.789$

IG(Humidity) = 0.940 - 0.789 = 0.151

4.4 Attribute: Wind

Weak subset -> rows 1, 3, 4, 5, 8, 9, 10, 13
 Yes = 6, No = 2
 Entropy(Weak) = $- (6/8) \log_2(6/8) - (2/8) \log_2(2/8)$
 $= - (0.75)(-0.415) - (0.25)(-2)$
 $= 0.3113 + 0.5000$
 $= 0.811$

Strong subset -> rows 2, 6, 7, 11, 12, 14
 Yes = 3, No = 3
 Entropy(Strong) = $- (3/6) \log_2(3/6) - (3/6) \log_2(3/6)$
 $= - (0.5)(-1) - (0.5)(-1)$
 $= 0.5 + 0.5$
 $= 1.000$

Weighted Entropy(Wind)
 $= (8/14)(0.811) + (6/14)(1.000)$
 $= 0.4634 + 0.4286$
 $= 0.8920$
 $= 0.892$

IG(Wind) = 0.940 - 0.892 = 0.048

5. Select the Root Node

Attribute	Information Gain
Outlook	0.246
Humidity	0.151

Attribute	Information Gain
Wind	0.048
Temperature	0.029

The maximum Information Gain is 0.246. Therefore, the root node is Outlook.

6. Continue Splitting the Branches

6.1 Branch: Outlook = Overcast

Rows: 3, 7, 12, 13
 Yes = 4, No = 0
 Entropy = 0
 So this is a pure node.

Overcast -> Yes

6.2 Branch: Outlook = Sunny

Sunny subset rows: 1, 2, 8, 9, 11
 Yes = 2, No = 3
 Entropy(Sunny) = 0.971

Now test the remaining attributes for the Sunny subset. Humidity gives a perfect split.

Humidity = High -> rows 1, 2, 8 -> Yes = 0, No = 3
 Entropy(High) = 0

Humidity = Normal -> rows 9, 11 -> Yes = 2, No = 0
 Entropy(Normal) = 0

Weighted Entropy(Sunny, Humidity)
 $= (3/5)(0) + (2/5)(0)$
 $= 0$

IG(Sunny, Humidity)
 $= \text{Entropy(Sunny)} - 0$
 $= 0.971$

So,
 Sunny + High -> No
 Sunny + Normal -> Yes

6.3 Branch: Outlook = Rain

Rain subset rows: 4, 5, 6, 10, 14
 Yes = 3, No = 2
 Entropy(Rain) = 0.971

Now test the remaining attributes for the Rain subset. Wind gives a perfect split.

Wind = Weak -> rows 4, 5, 10 -> Yes = 3, No = 0
 Entropy(Weak) = 0

Wind = Strong -> rows 6, 14 -> Yes = 0, No = 2
 Entropy(Strong) = 0

Weighted Entropy(Rain, Wind)
 $= (3/5)(0) + (2/5)(0)$
 $= 0$

IG(Rain, Wind)

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= Entropy(Rain) - 0
= 0.971
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So,
Rain + Weak -> Yes
Rain + Strong -> No

7. Final Decision Tree

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Outlook
|-- Overcast -> Yes
|-- Sunny
|   |-- Humidity
|       |-- High -> No
|       |-- Normal -> Yes
|-- Rain
    |-- Wind
        |-- Weak -> Yes
        |-- Strong -> No
```

8. Final Rules

1. If Outlook = Overcast, then PlayTennis = Yes
2. If Outlook = Sunny and Humidity = High, then PlayTennis = No
3. If Outlook = Sunny and Humidity = Normal, then PlayTennis = Yes
4. If Outlook = Rain and Wind = Weak, then PlayTennis = Yes
5. If Outlook = Rain and Wind = Strong, then PlayTennis = No

Classroom note: in this dataset, both Entropy and Gini produce the same final tree, but the measures used to choose the split are different.